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End Semester Examination 2025

Name of the Course: B. Tech.

Semester: 5th

Name of the Paper Geotechnical and Foundation Engineering

Paper Code: **TCE 503**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Write a Short note on Bore logs explain with the help of an example of bore log.	CO1
(b)	Explain the importance of soil exploration program. Explain two methods of subsurface exploration	
(c)	Explain the following Geophysical Tests (Seismic Refraction Method, Electrical Resistivity Method).	
Q2	(20 marks)	
(a)	A retaining wall, 6 m high, retains dry sand with an angle of friction of 30° and unit weight of 16.2 kN/m^3 . Determine the earth pressure. If the water table rises to the top of the wall, determine the increase in the thrust on the wall. Assume the submerged unit weight of sand as 10 kN/m^3 . Draw pressure distribution diagram and show the force in it.	CO2
(b)	A retaining wall, 8 m high, retains a cohesionless backfill. The top 4 m of the fill has a unit weight of 18 kN/m^3 and $\phi = 30^\circ$ and the rest has unit weight of 24 kN/m^3 and $\phi = 20^\circ$. Determine the active and passive pressure distribution on the wall. Make pressure diagram also.	
(c)	Explain the following terms related to earth retaining structures along with the help of diagram. Sheet pile wall and its types, Anchored Bulkhead, Cofferdams, and Braced Cuts.	
Q3	(20 marks)	
(a)	Enlist assumptions of Terzaghi theory. Explain different types of shear failure of shallow foundation.	CO3
(b)	Compute the safe bearing capacity of a continuous footing 1.8 m wide, and located at a depth of 1.2 m below ground level in a soil with unit weight $\gamma = 20 \text{ kN/m}^3$, $c = 20 \text{ kN/m}^2$, and $\phi = 20^\circ$. Assume a factor of safety of 2.5. Terzaghi's bearing capacity factors for $\phi = 20^\circ$ are $N_c = 17.7$, $N_q = 7.4$, and $N_\gamma = 5.0$, what is the permissible load per metre run of the footing?	
(c)	In a borehole, a layer of fine sand was encountered at a depth of 5m from the ground surface. A dense sand layer of density 16 kN/m^3 overlies the fine sand layer of density 14 kN/m^3 . The Average N value recorded in the field was 23. Determine the correct N value for the layer at 5m depth. The water table was found to be at a depth of 2m from surface. Determine the bearing capacity of a square footing 2.0 m wide and use Peck Henson's equation.	
Q4	(20 marks)	
(a)	A group of 16 piles of 50 cm diameter is arranged with a centre to centre spacing of 1.0 m. The piles are 9 m long and are embedded in soft clay with cohesion 30 kN/m^2 . Bearing resistance may be neglected for the piles—	CO4

	Adhesion factor is 0.6. Determine the ultimate load capacity of the pile group.	
(b)	Design a pile group to carry 400 kN in clay with a UCS of 60 kN/m ² . Adhesion factor may be taken as 0.6. The piles are of size 300mm and 6m long. Assume FOS as 2.5.	
(c)	Explain the necessity of using a pile foundation. Classify the pile foundations (only name) on the behalf of function and method of installation with the help of neat diagram.	
Q5	(20 marks)	
(a)	Derive an expression for FOS for stability of soil slopes in infinite C- ϕ deposits with neat diagram.	CO5
(b)	An embankment is inclined at an angle of 35° and its height is 15 m. The angle of shearing resistance is 15° and the cohesion intercept is 200 kN/m ² . The unit weight of soil is 18.0 kN/m ³ . If Taylor's stability number is 0.06, find the factor of safety with respect to cohesion.	
(c)	A long Natural Slope in a clayey deposit having $c = 10 \text{ kN/m}^2$, $\phi = 25^\circ$, $\gamma_{\text{Sat}} = 20 \text{ kN/m}^3$ is inclined at 10° to the horizontal. The water table is at the surface and seepage is parallel to the slope. If a plane slip had developed at a depth of 5m below the surface, determine the FOS of slope. Take $\gamma_w = 10 \text{ kN/m}^3$	